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Report Title: SEG3125 Assignment 3 — Memory Game Page Report

Game

a. **Name and Type of Game**

The game I decided to develop is called MN Powerlifting Technique Memory Game. It is a powerlifting-themed matching card memory game. The user flips cards to find matching pairs while reviewing squat, bench press, and deadlift technique cues. The goal of the game is to help users practice their memory by remembering the position of each card and matching the same lifting cue.

The game includes multiple levels, starting from Level 0 with 6 cards and increasing up to Level 5 with 30 cards. It also includes a configuration option called Display Mode, where users can choose between Image + Cue Mode and Cue Focus Mode. In Image + Cue Mode, the card shows a lifting image with a technique cue. In Cue Focus Mode, the card shows only the cue on a red background, so the user can focus more directly on memorizing the text.

b. **Inspiration Sites**

Inspiration Site 1: Memory: Play Free Online – Brain Training

This site inspired the basic matching-card structure of my game. In the example, users flip cards in a grid and try to find matching pairs. I used this idea as the foundation for my own game, but changed the theme to powerlifting. Instead of using general images, my game uses squat, bench press, and deadlift technique cues. I also improved the design by adding a stronger visual theme, clearer feedback, multiple levels, and a powerlifting identity.

Inspiration Site 2: https://quizlet.com/ca

Quizlet inspired the second storyboard of my memory game. In Quizlet, users can study knowledge points through flashcards and then test themselves using quiz-style activities. I used this idea to design a cue memory quiz storyboard, where users first review SBD technique cues and then answer multiple-choice questions based on what they remember. This inspired the “Cue Focus” direction of my final prototype, which removes images and helps users focus more directly on memorizing the technique text.

Persona

AMY

Amy is a beginner lifter who already has some basic experience with strength training.
She is interested in entering the field of powerlifting, but she does not yet feel confident about her squat, bench press, and deadlift technique.

INTRINSIC CHARACTERISTICS

- **Beginner powerlifting learner**
- **Motivated and curious**
- **Sometimes forgets technical cues easily**

RELATION TO TECHNOLOGY

- Comfortable using websites and simple games:
- Likes interactive and visual learning tools

RELATION TO THE DOMAIN

- New to squat, bench press, and deadlift
- Wants to learn the basic cues correctly

GOAL

Amy wants to connect each squat, bench press, and deadlift image with an important technique cue. By playing the game, she hopes to remember basic lifting reminders in a more visual and interactive way, so she can feel more confident when practicing in the gym.

INTRINSIC CHARACTERISTICS

- **Intermediate lifter**
- **Goal-oriented**
- **Busy and practical**

RELATION TO TECHNOLOGY

- Uses fitness apps and training tools regularly
- Prefers efficient and straightforward interfaces

RELATION TO THE DOMAIN

- Already knows SBD basics
- Wants a fast way to review important cues before training

GOAL

Ryan wants to use the memory game as a short review tool before training. He hopes the higher levels can challenge his memory, help him focus on important cues, and remind him of small technique details before heavy squat, bench press, or deadlift sessions.

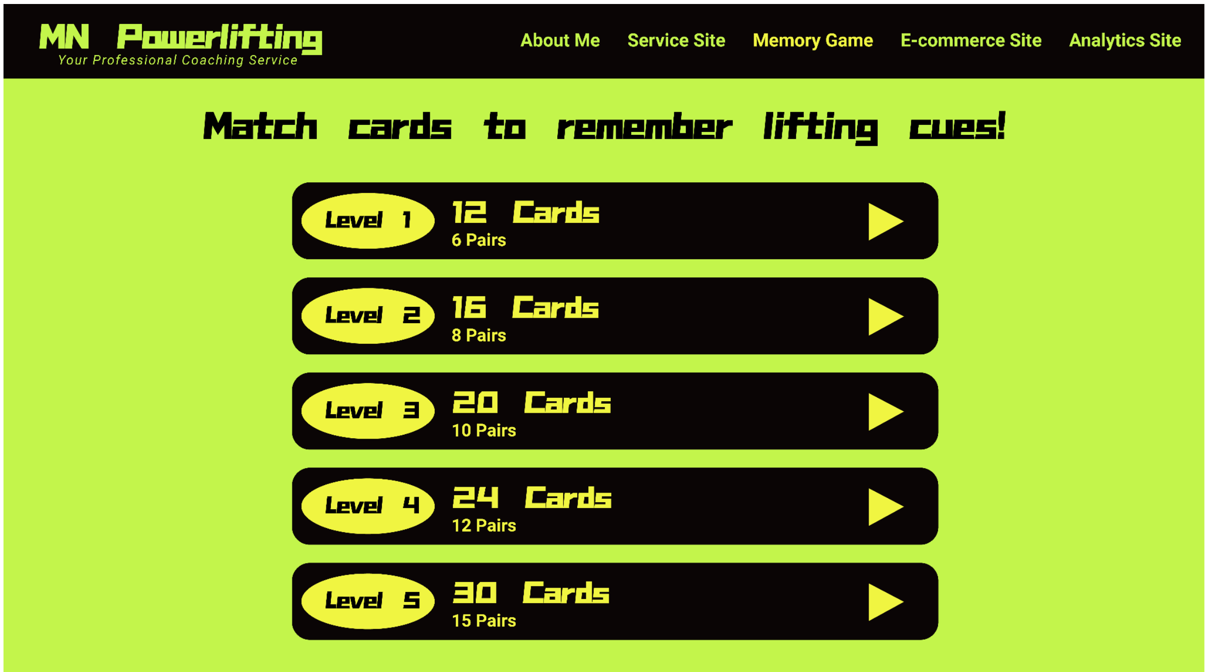
RYAN

Ryan is an experienced powerlifter with 5 years of powerlifting training experience. He has competed in 3 powerlifting competitions and has used this coaching service before. He is now preparing for another competition and wants a more structured training plan.

Storyboard

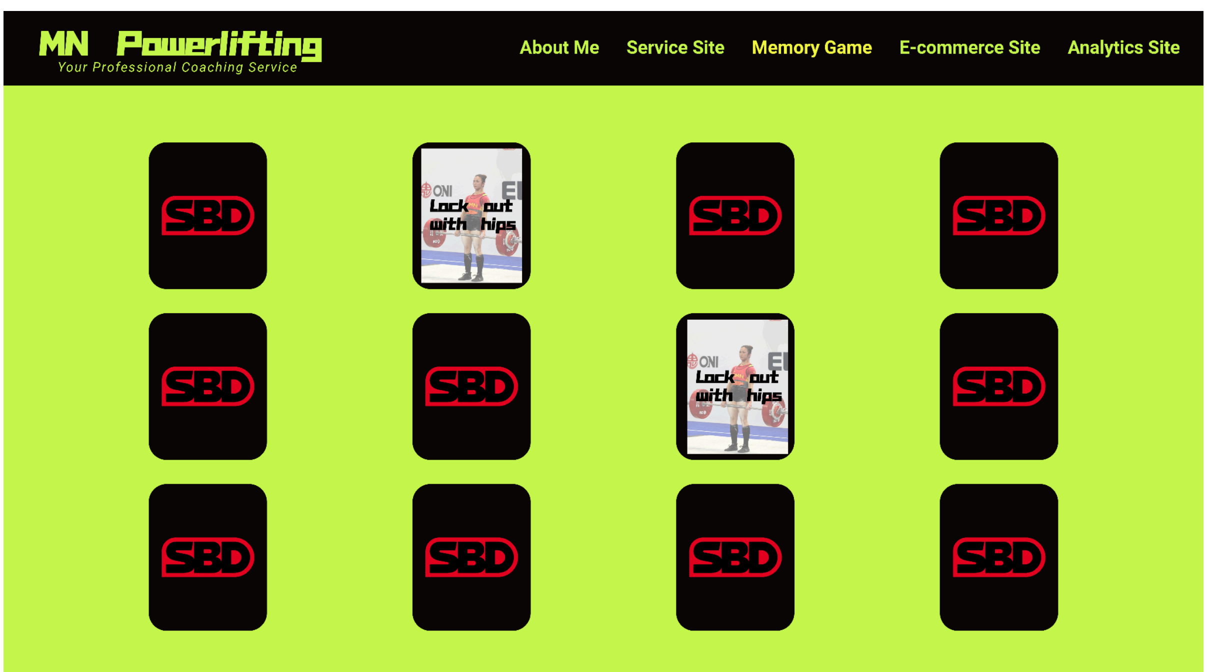
1. **Amy's Storyboard**
   Step 1: Level Selection

Amy opens the MN Powerlifting Technique Memory Game and views the available levels. Since she is a beginner, she chooses a lower level with fewer cards so she can start practicing the lifting cues more easily.

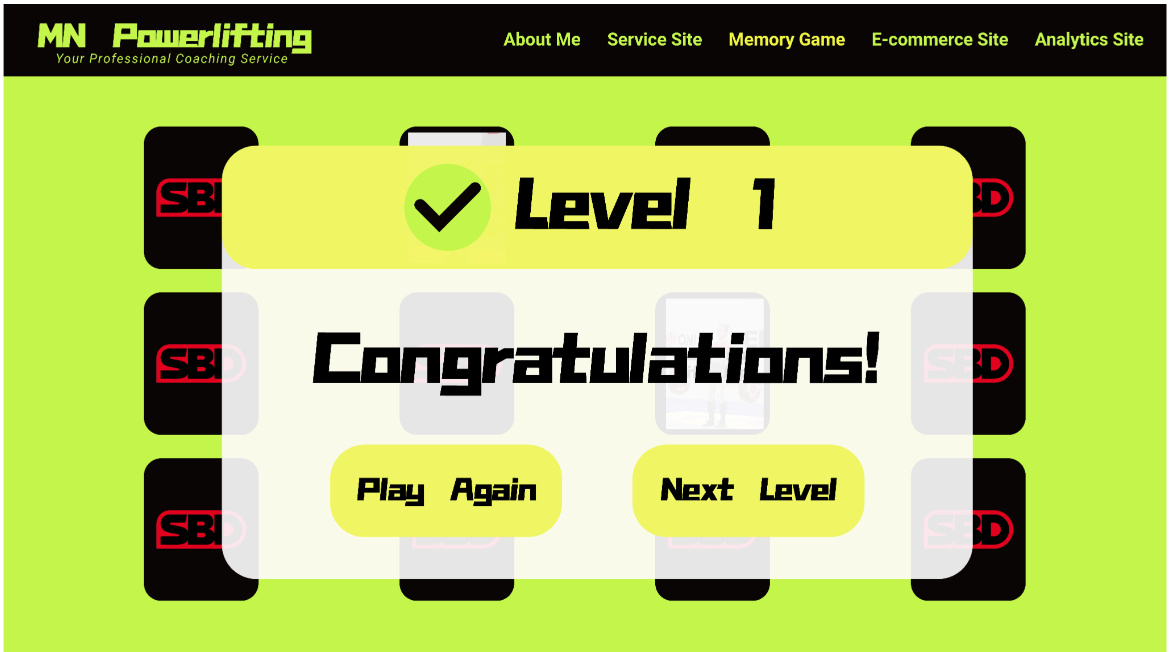


Step 2: Playing the Memory Game

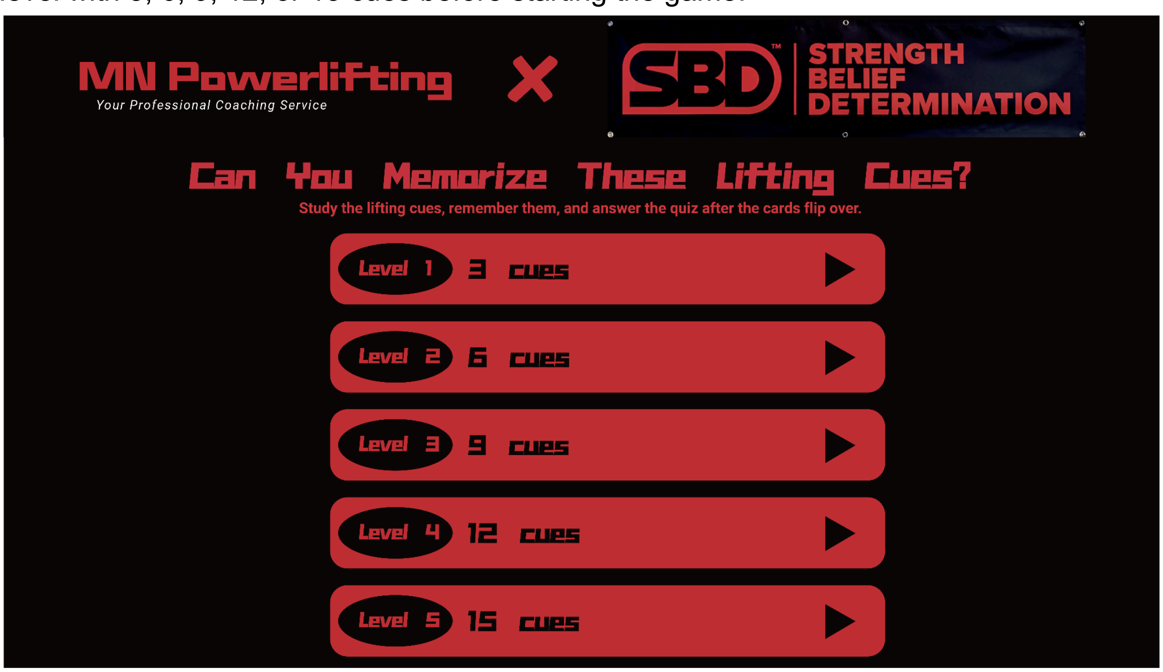
Amy starts the game and flips two cards at a time. Each card shows an SBD image with a technique cue. She tries to remember the position of each cue and match the same cards together.



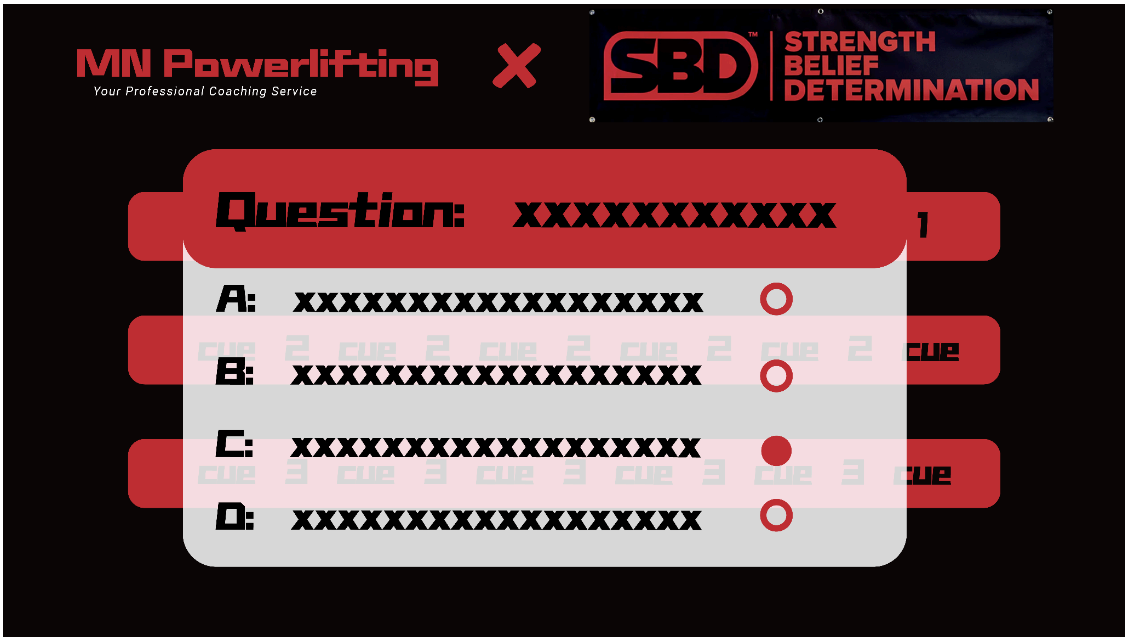
Step 3: Game Completion
After Amy matches all the cards, a completion screen appears. It shows a congratulations message and tells her that she has finished the level. She can choose to play again or move to the next level.



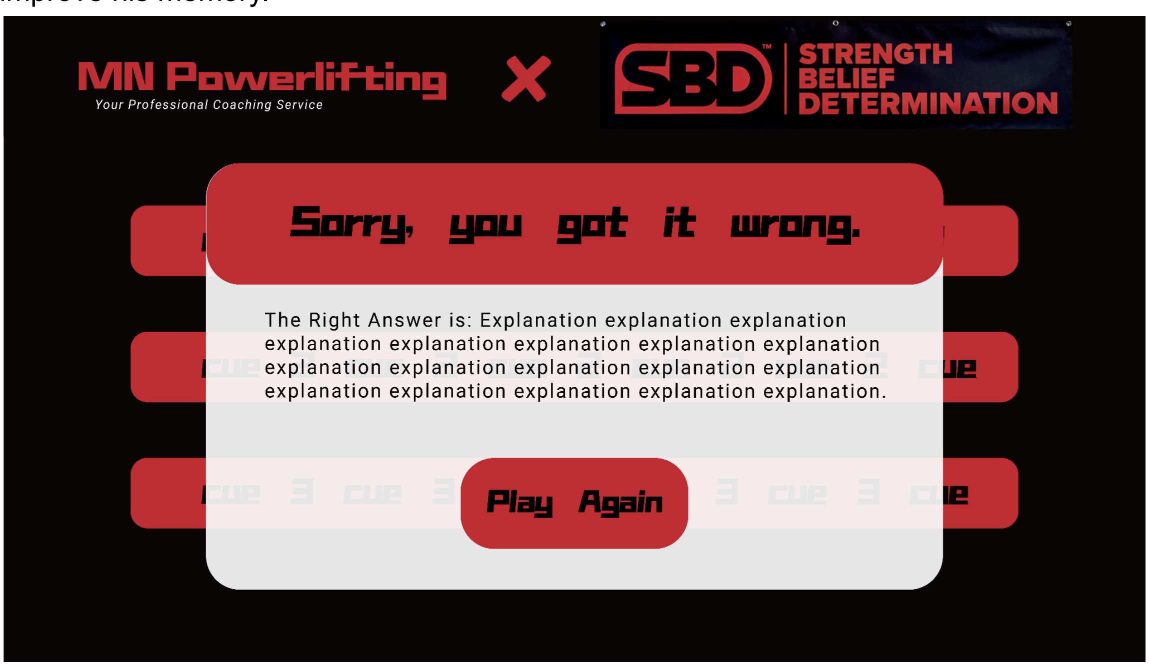
2. **Ryan's Storyboard**
   Step 1: Level and Cue Selection
   Ryan opens the quiz-style SBD memory game and chooses a level based on how many lifting cues he wants to memorize. Since he wants a quick challenge, he can select a level with 3, 6, 9, 12, or 15 cues before starting the game.



Step 2: Memory Quiz with Feedback
After studying the lifting cues, the cards flip over and Ryan answers a multiple-choice question based on what he remembers. He selects one answer from the options, and the interface gives immediate feedback by showing whether the selected answer is correct or incorrect.



Step 3: Game Completion
After Ryan answers the quiz question, the game shows a result screen. If the answer is wrong, the page displays a short message such as “Sorry, you got it wrong,” followed by the correct answer or explanation. Ryan can then choose to play again and try to improve his memory.



Visual Design Choice
For the high-fidelity prototype, I used a black, red, and white color theme to match the MN Powerlifting brand and create a strong powerlifting atmosphere. Black is used for the navigation bar and card backs, red is used for buttons, highlights, and cue focus cards, while white is used as the main background to keep the interface clean and readable.

The typography uses bold display fonts to create an athletic and energetic feeling. Large headings and level numbers help attract attention, while smaller text such as feedback, moves, and matches stays simple and easy to read.

The layout is designed to be clear and task-focused. The level selection page uses large horizontal rows so users can compare levels easily. The game page uses a card grid, which helps users remember card positions. The end screen appears as a semi-transparent overlay, so users receive feedback without leaving the game page.

Negative space is used between level rows, cards, and sections to avoid visual clutter. This makes each option and card easier to recognize.

Several Gestalt principles are also applied. Similarity is shown through cards with the same size and shape. Proximity groups the level number, card count, and play button together. Figure-ground is created through strong contrast between the black cards, red highlights, and white background. Common region is used in the end-game overlay, where the result message and buttons are grouped in one panel.

Link to UI
ziye-seg3125-project.netlify.app

Link to GitHub Repository
https://github.com/Ziye-Zhou/SEG3125_Project

Generative AI Acknowledgement
I used generative AI as a support tool during this assignment, but the role of AI was different for each part of the project.

For the mockups and storyboards, I did not use AI to create the visual design directly. I created the layouts, color themes, typography choices, and overall visual style myself using design tools. Since I have personal knowledge of powerlifting, I also developed the main SBD technique cue idea myself. However, AI helped me organize the storyboard structure by suggesting how to divide each persona's goal into clear steps, including level selection, gameplay with feedback, and the end-of-game screen.

For the high-fidelity prototype, AI was most helpful with the React development. I used AI to help plan and revise the interactive structure of the memory game, including the level system, display mode configuration, card-flipping logic, matching feedback, score tracking, and end-game overlay. AI also helped me debug issues when converting the prototype from regular JavaScript to a React-based version and when fixing problems related to script loading, layout, and styling. I tested and adjusted the code myself to make sure the final prototype matched my intended design and worked correctly.

For the report and submission process, AI helped me organize my explanations and make the writing clearer. It also helped me describe the visual design choices, Gestalt principles, personas, storyboard steps, and the role of React in the prototype. The final design decisions, wording, testing, and submitted work were reviewed and edited by me.